

Spring AI Gateway + Ollama + WhatsApp Bot Setup Guide

This guide explains how to create and run a Spring Boot AI Gateway using: Java 21 Spring Boot Spring AI Ollama Qwen2.5 model YAML configuration WhatsApp Bot integration

1. Recommended Architecture

```
WhatsApp Bot
  ↓
Spring AI Gateway
  ↓
Ollama
  ↓
qwen2.5:1.5b
```

2. Why Use YAML

YAML is better for:

- Spring AI
- Docker
- Kubernetes
- Microservices
- Nested configuration

Example:

```
spring:
  ai:
    ollama:
      chat:
        model: qwen2.5:1.5b
```

3. Create Spring Boot Project

Open Spring Initializr:
<https://start.spring.io>

Select:

- Project: Maven
- Language: Java
- Spring Boot: Latest Stable
- Java: 21
- Packaging: Jar

Dependencies:

- Spring Web
- Lombok
- Spring AI Ollama

Configuration Format:

- YAML

4. Download and Extract Project

Commands:

```
cd ~/Desktop/sameer-ai-platform/services
```

```
unzip ~/Downloads/ai-gateway.zip
```

```
If asked:  
replace ...
```

```
Press:  
A
```

```
Verify:
```

```
cd ai-gateway  
ls
```

```
Expected:  
HELP.md  
mvnw  
mvnw.cmd  
pom.xml  
src
```

5. Run Spring Boot Application

```
Go into project:
```

```
cd ~/Desktop/sameer-ai-platform/services/ai-gateway
```

```
Run:
```

```
mvn spring-boot:run
```

```
Expected Output:
```

```
Started AiGatewayApplication  
Tomcat started on port 8080
```

6. Fix Missing Spring AI Dependency

```
If build fails, open:
```

```
pom.xml
```

```
Add:
```

```
<dependency>  
  <groupId>org.springframework.ai</groupId>  
  <artifactId>spring-ai-starter-model-ollama</artifactId>  
</dependency>
```

7. Configure application.yml

```
Create or edit:
```

```
src/main/resources/application.yml
```

```
Example:
```

```
spring:  
  ai:  
    ollama:  
      base-url: http://localhost:11434  
      chat:  
        model: qwen2.5:1.5b
```

```
server:  
  port: 8080
```

8. Install and Run Ollama

```
Install Ollama:
curl -fsSL https://ollama.com/install.sh | sh

Run model:
ollama run qwen2.5:1.5b

Verify:
ollama list
```

9. Create AI Chat API

```
Create:
src/main/java/com/example/aigateway/controller/AiController.java

Example endpoint:
POST /api/ai/chat

This endpoint will connect:
Spring AI → Ollama → Qwen Model
```

10. Connect WhatsApp Service

```
Connect your WhatsApp bot service to:
http://localhost:8080/api/ai/chat

Flow:
WhatsApp User
  ↓
WhatsApp Service
  ↓
Spring AI Gateway
  ↓
Ollama
  ↓
Qwen2.5 Model
```

11. Useful Commands

```
Check Java:
java -version

Check Maven:
mvn -version

Run App:
mvn spring-boot:run

Stop App:
CTRL + C

Check Ollama:
```

ollama list

12. Recommended Future Stack

Always use:

- application.yml
- Spring AI
- Docker
- Kubernetes
- Microservices architecture
- Ollama local models

Recommended for scalable AI systems.

Final Notes

You are now ready to build:

WhatsApp → Spring AI → Ollama → qwen2.5

This architecture can later expand into:

- AI agents
- RAG systems
- Vector databases
- Multi-model AI systems
- Kubernetes deployments
- Cloud-native microservices